Gopalganj Science and Technology University

Department of Electrical and Electronic Engineering

Application for the Approval of Thesis Proposal

Minimization of Leakage Current and Frequency Range Determination in Transformer-less Inverters for Photovoltaic On Grid Systems.

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CERTIFICATION

This is to certify that the thesis titled:

“Minimization of Leakage Current by Frequency Range Determination & Increasing Thermal Efficiency in Transformer-less Inverters for Photovoltaic On-Grid Systems”

submitted by the following students:

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Inpartial fulfillment of the requirements for the degree of **Bachelor of Science in Electrical and Electronic Engineering**, has been carried out under my supervision.

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Minimization of Leakage Current by Frequency Range Determination & Increasing Thermal Efficiency in Transformer-less Inverters for Photovoltaic on Grid Systems.

A B S T R A C T

This study examines ways to improve the efficiency and safety of solar inverters. Because these inverters lack a transformer, they frequently produce undesired electric current known as "leakage current" when utilized in solar power systems. Performance may suffer and safety issues may arise. Finding the optimal switching frequency and inverter design to lower this current is the aim of this investigation. The study tests various inverter kinds and temperatures using computer models. The findings will aid in selecting the most cost-effective, safe, and effective solar energy system design.

INTRODUCTION

In modern grid-connected solar power systems, transformer-less inverters have gained popularity due to their compact size, higher efficiency, lower cost, and ease of maintenance. However, the elimination of the transformer leads to the absence of galvanic isolation between the PV array and the grid. This causes leakage current, mainly due to common-mode voltage variation across the parasitic capacitance of the PV system.

- This leakage current can cause:
- Electromagnetic interference (EMI)
- Unnecessary tripping of protection devices
- Reduced inverter lifespan
- Safety hazards

Therefore, better efficiency, effective control for leakage current reduction is essential. Leakage current in PV systems arises from parasitic capacitance $(C_{PV} - GND)$ between the array and ground, caused by the dielectric properties of the panel layers. It is calculated as $C = \epsilon \times \frac{A}{dC} = \epsilon \times \frac{A}{d}\ and$ generates leakage current $Ileakage = C \times \left(\frac{d}{dt} V_{CM}\right)$ during inverter switching, leading to EMI, safety risks, and insulation degradation.
There are few methods to reduce leakage current.

- Use of Specific Inverter Topologies
- Common Mode Voltage (CMV) Control
- Isolation and Filtering
- Grounding Schemes
- Software-Based Control Strategies

LITERATURE & REVIEW

Transformer-less inverters are preferred in grid-connected PV systems due to higher efficiency, lower cost, and compact size. However, leakage current remains a key challenge due to the lack of galvanic isolation.

Key Findings from Literature:

1. Schmidt et al. (2005) – Identified leakage current issues caused by common-mode voltage fluctuations and suggested topological improvements.
2. Kjaer, Pedersen, Blaabjerg (2005) – Compared inverter topologies and recommended optimal designs for transformer-less systems.
3. H5 & H6 Topologies – Introduced additional switches (H5 by Victor Vinnikov, H6 by Y. Yang) to decouple PV from the grid during freewheeling, reducing leakage current.
4. HERIC Inverter (2007) – Used auxiliary switches to stabilize common-mode voltage, effectively minimizing leakage.
5. Modulation Techniques – Zero Common Mode PWM and Unipolar PWM helped maintain constant/zero common-mode voltage.
6. Recent Advances (2020–2024) – Focus on intelligent control, soft-switching, and AI-based optimization for further leakage suppression.

SCOPE OF THE WORK

The scope encompasses theoretical analysis, simulation, and performance validation of transformer-less inverters for low-to-medium power (1–5 kW) PV systems. The work includes:

1. Simulation of Topologies:
   - Model HERIC, H5, H6, and NPC configurations in MATLAB/Simulink.
   - Implement PWM strategies (sinusoidal, hybrid modulation) and dead-time compensation.
2. Leakage Current Analysis:
   - Measure CMV variations and parasitic capacitance effects.
   - Use time-domain simulations and FFT to analyze leakage current harmonics.
3. Frequency Optimization:
   - Sweep switching frequencies (5–30 kHz) to establish efficiency-leakage trade-offs.
   - Apply Pareto optimization to identify the ideal frequency range.
4. Thermal Stress Assessment:
   - Simulate power losses (conduction, switching) and thermal profiles under varying loads.
   - Propose heat dissipation methods (e.g., heat sinks, airflow design).
5. Standards Compliance:

- Validate leakage current levels against IEC 62109 (<300 mA) and grid interconnection norms (IEEE 1547).

6. Practical Guidelines:
   - Provide recommendations for topology selection and cooling systems in real-world applications.

OBJECTIVES

The primary objectives of this thesis are to design, analyze, and optimize transformer-less inverters for grid-connected photovoltaic (PV) systems, focusing on suppressing leakage current and enhancing thermal efficiency. The study aims to:

- ✓ Model and simulate key transformer-less topologies **(HERIC, H5, H6, NPC)** to evaluate their leakage current performance.
- ✓ Investigate the impact of switching frequency (5–30 kHz) on leakage current magnitude, electromagnetic interference (EMI), and inverter efficiency.
- ✓ Identify an optimal frequency range that minimizes leakage current while maintaining thermal stability and compliance with IEC 62109 safety standards.
- ✓ Compare topologies based on efficiency, total harmonic distortion (THD), and leakage suppression capabilities.
- ✓ Analyze thermal stress on semiconductor devices **(IGBTs/MOSFETs)** to recommend cooling strategies and improve reliability.

METHODOLOGY

This study investigates leakage current mechanisms in transformer-less PV inverters, focusing on the HERIC topology and comparative analysis with other topologies (H5, NPC, Four Switch). The methodology consists of five key phases, combining simulation, thermal analysis, and performance evaluation.

1. Literature Review & Theoretical Analysis
   - Review existing research on transformer-less inverter topologies and leakage current mechanisms.
   - Analyze common-mode voltage (CMV) variations and their impact on leakage current.

2. Circuit Design & Simulation in MATLAB/Simulink
   - Model HERIC, H5, and H6 topologies with precise switching logic.
   - Implement PWM control strategies (sinusoidal PWM, hybrid modulation).
   - Incorporate dead-time compensation to minimize distortion.
   - Design freewheeling path control to stabilize CMV.

3. Leakage Current Simulation & Performance Evaluation

- Simulate leakage current under varying switching frequencies (5 kHz – 30 kHz).
- Measure leakage current magnitude and inverter efficiency.

4. Thermal Analysis Using Simscape
   - Model power loss distribution in semiconductor devices (IGBTs/MOSFETs/SCRs).
   - Conduct thermal stress analysis under different load conditions.
5. Result Validation & Compliance Check
   - Compare leakage current levels with IEC 62109 safety standards.
   - Identify optimal switching frequency range for efficiency & safety.
   - Provide recommendations for topology selection based on performance metrics.

This structured approach ensures a comprehensive evaluation of leakage current mitigation techniques while maintaining high efficiency in transformer-less PV inverters.

EXPECTED OUTCOMES

The research is anticipated to yield the following practical and theoretical outcomes, directly addressing leakage current minimization and thermal efficiency in transformerless PV inverters:

1. Optimized HERIC Inverter Model:

- A validated MATLAB/Simulink model of the HERIC topology with active CMV stabilization, achieving leakage current reduction by >50% compared to conventional H-bridge designs.
- Implementation of hybrid PWM modulation to minimize dead-time distortion and improve waveform quality (THD <3%).

2. Switching Frequency-Leakage Current Relationship:

- Graphical plots (e.g., leakage current vs. frequency, efficiency vs. frequency) illustrating how 5–30 kHz switching frequencies impact performance.
- Identification of a sweet-spot range (e.g., 10–20 kHz) where leakage current is suppressed (<150 mA) without compromising efficiency (>97%).

3. Thermal Profiles and Cooling Recommendations:

- Thermal maps showing junction temperatures of IGBTs/MOSFETs under varying loads (25°C–50°C ambient).
- Guidelines for heat sink design and airflow management to extend device lifespan by 15–20%.

4. Topology Comparison Framework:

- A quantitative comparison table ranking HERIC, H5, H6, and NPC topologies based on:
- Leakage current (HERIC expected to perform best).
- Efficiency (H6 for low-loss applications).
- Component count (NPC for multilevel voltage output).

5. Compliance with Safety Standards:
   - Validation of leakage current levels below 300 mA (IEC 62109) and THD <5% (IEEE 1547), ensuring grid compatibility.

6. Practical Implementation Guidelines:
   - A decision matrix for engineers to select topologies based on cost, efficiency, and application scale (1–5 kW systems).
   - Recommendations for soft-switching integration or AI-driven modulation to further reduce losses in future work.

CONCLUSION

The increasing adoption of transformer-less inverters in grid-connected photovoltaic (PV) systems underscores the need to address persistent challenges such as leakage current, electromagnetic interference (EMI), and thermal inefficiency. This thesis proposal systematically investigates the minimization of leakage current through optimized switching frequency ranges and topology selection, focusing on **HERIC**, **H5**, and **H6** inverter configurations. By leveraging **MATLAB/Simulink** simulations and thermal analysis in **Simscape**, the study evaluates the interplay between switching frequency, leakage current magnitude, and semiconductor device stress.

Key expected outcomes include the identification of an optimal switching frequency range **(5–30 kHz)** that balances leakage current suppression with thermal stability and inverter efficiency. Comparative analysis of topologies will highlight the HERIC configuration's advantages in common-mode voltage (CMV) stabilization, while thermal simulations will provide insights into power loss distribution across **IGBTs/MOSFETs**. Compliance with **IEC 62109** standards will validate the safety and reliability of the proposed solutions.

This research contributes practical guidelines for designing efficient, low-leakage transformer-less inverters in low-to-medium power PV systems **(1–5 kW).** Future work could explore AI-driven modulation techniques, hardware prototyping, and scalability for higher power applications. By advancing leakage current mitigation strategies, this study supports the broader goal of enhancing the sustainability and safety of renewable energy integration into modern grids.

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